

2 Prove by mathematical induction that, for all positive integers n ,

$$\frac{d^n}{dx^n}(x^2 e^x) = (x^2 + 2nx + n(n-1))e^x. \quad [6]$$

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for writing. The lines are evenly spaced across the entire page, providing a guide for letter height and placement. There is no text or other markings on the page.